

Transfer Learning

Deep Learning Assignment Report

Submitted by

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1. Introduction

Transfer learning is a machine learning technique where a model trained on a large source task is reused as the starting point for a different but related target task. Instead of training a deep neural network from scratch - which requires millions of labelled examples and significant compute - transfer learning leverages representations already learned from large datasets such as ImageNet (1.2 million images, 1000 classes).

CNNs trained on ImageNet learn a rich hierarchy of visual features: early layers capture edges and color gradients, middle layers encode textures and shapes, and later layers represent high-level semantic concepts. These representations generalize well across many visual tasks.

This report investigates transfer learning using ResNet50, a 50-layer residual network pretrained on ImageNet, applied to the Oxford Flowers102 dataset - a fine-grained classification benchmark with 102 flower categories. Six distinct freezing strategies are evaluated, ranging from total feature extraction to gradual unfreezing (ULMFiT). The results demonstrate the critical interplay between dataset size, layer freezing strategy, and catastrophic forgetting.

2. Dataset and Model Architecture

Oxford Flowers102 Dataset

The Oxford Flowers102 dataset consists of 102 flower categories. Only ~10 training images per class therefore an ideal benchmark for evaluating transfer learning under extreme data scarcity.

Split	Samples	Classes	Per Class
Train	1,020	102	10
Validation	1,020	102	10
Test	6,149	102	~60

ResNet50 Architecture

ResNet50 has around 25.6M parameters that uses skip connections to enable deep training without vanishing gradients. For all experiments, the ImageNet head was replaced with: GlobalAveragePooling2D → Dense (256, ReLU) → Dropout (0.2) → Dense (102, Softmax).

Layer Group	Layers	Features Learned	Frozen In Experiments
conv1+bn1	0–4	Edges, colour	B, C, D, E
conv2_x	5–36	Shapes, blobs	B, C, D, E
conv3_x	37–78	Textures, patterns	B, C, D
conv4_x	79–141	Object parts	B, C
conv5_x	142–174	High-level semantics	B only

3. Experiments

Six experiments were conducted, each using the same backbone and custom head but differing in which layers were trainable. All used Adam optimiser with sparse categorical cross-entropy loss. Early stopping (patience=10, restore_best_weights=True) and ReduceLROnPlateau (factor=0.5, patience=3) were applied to B–D. Experiments E and F used early stopping only.

Experiment A - Total Freeze (Evaluation Only)

- ❖ All layers including the head were frozen.
- ❖ No training performed. Establishes a random-head baseline.
- ❖ Result: **0.33% test accuracy** - confirms an untrained head produces random outputs.

Experiment B - Feature Extraction

- ❖ Entire backbone frozen.
- ❖ Only the custom head (~551K params, 2.3%) trained. LR=1e-3, max 50 epochs.
- ❖ Result: **26.51% test accuracy**.
- ❖ The large jump from A proves ImageNet features transfer meaningfully even to a fine-grained flower task.

Experiment C – Partial Unfreezing (conv5_x + head)

- ❖ Layers 0–141 frozen, conv5_x (layers 142+) unfrozen. LR=1e-3, max 50 epochs.
- ❖ Result: **34.25% test accuracy**
- ❖ ~8% gain over B confirms that the highest-level backbone features needed domain-specific adaptation.

Experiment D – Fine-Tuning (conv4_x, conv5_x + head)

- ❖ Layers 0–78 frozen, conv4_x and conv5_x unfrozen (~22.6M params). LR=1e-4, max 50 epochs.
- ❖ Result: **47.28% test accuracy**
- ❖ Best among static strategies.
- ❖ Lower LR critical to protect pretrained weights.

Experiment E – Full Unfreezing

- ❖ All 24M+ params trained simultaneously.
- ❖ Fresh model, LR=1e-4, max 50 epochs, no ReduceLROnPlateau.
- ❖ Result: **23.74% test accuracy**
- ❖ Catastrophic forgetting observed.
- ❖ Val accuracy collapsed from ~34% to <3% by epoch 6.
- ❖ Demonstrates that full unfreeze on 1020 samples is unsustainable.
- ❖ Gradient updates from 10 images/class cannot guide 24M parameters coherently.

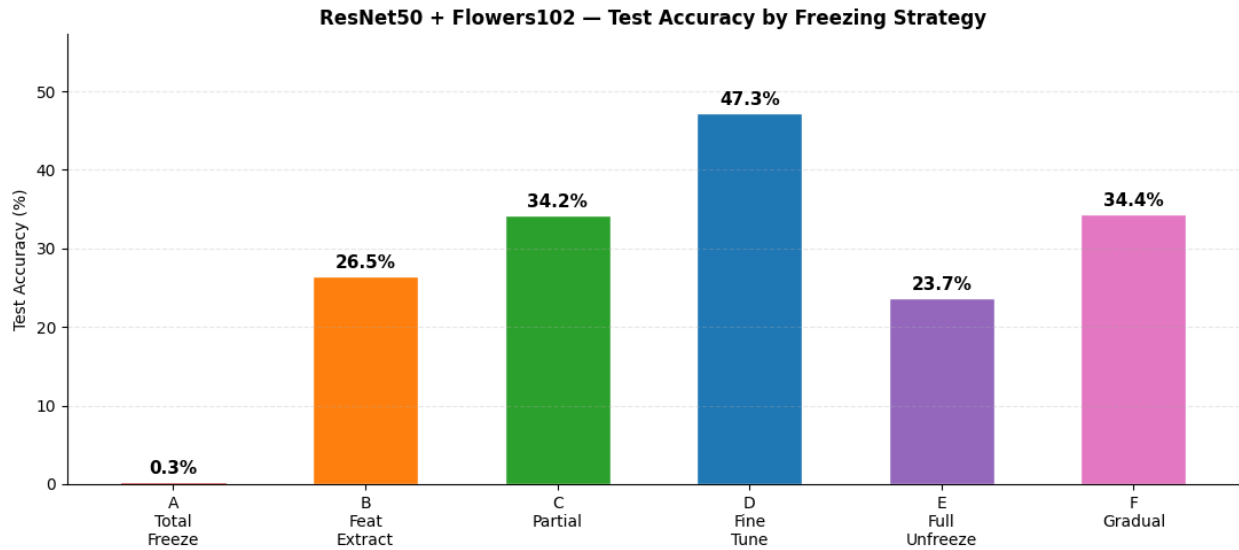
Experiment F – Gradual Unfreezing

- ❖ Phases: (1) Head only at lr=1e-2, (2) +conv5_x at lr=1e-3, (3) +conv4_x at lr=1e-4. 20 epochs per phase.
- ❖ Phase 4 (conv3_x) omitted - preliminary tests showed immediate collapse on this dataset size.
- ❖ Result: **34.41% test accuracy**
- ❖ Significantly better than Exp E (23.74%), validating the gradual approach.
- ❖ Phase-by-phase release prevents destructive interference between newly-unfrozen layers and previously-adapted ones.

4. Results and Analysis

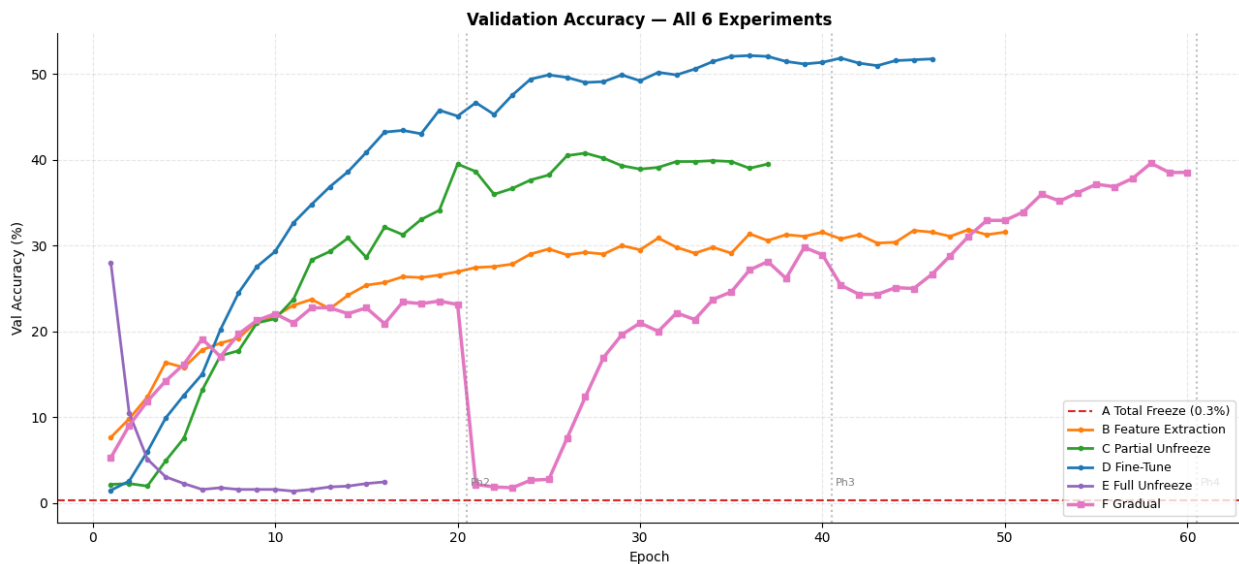
Summary Table

Exp	Strategy	Trainable	Epochs	Test Acc	Key Finding
A	Total Freeze (Eval)	0	-	0.33%	Random - untrained head
B	Feature Extraction	551K (2.3%)	50	26.51%	ImageNet features transfer
C	Partial Unfreeze	5.2M (20%)	37	34.25%	conv5 adapts to flowers
D	Fine-Tune (conv4+5)	22.6M (88%)	46	47.28%	Best result - partial freeze
E	Full Unfreeze	24.1M (99%)	16	23.74%	Catastrophic forgetting
F	Gradual Unfreeze	22.6M (88%)	60	34.41%	Stable - prevents collapse



Test Accuracy (%) by Freezing Strategy - ResNet50 + Flowers102

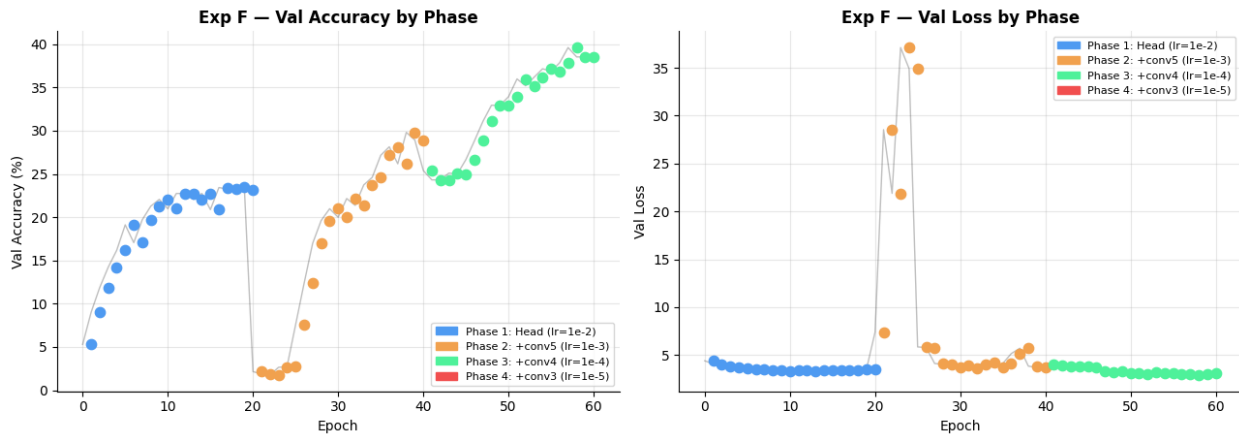
The progressive improvement from A (0.33%) through B→C→D (26%→34%→47%) confirms that incremental unfreezing of deeper layers improves performance. Notably, Experiment E (full unfreeze, 23.74%) underperforms even B, demonstrating catastrophic forgetting on a tiny dataset.



Validation Accuracy Learning Curves - All 6 Experiments

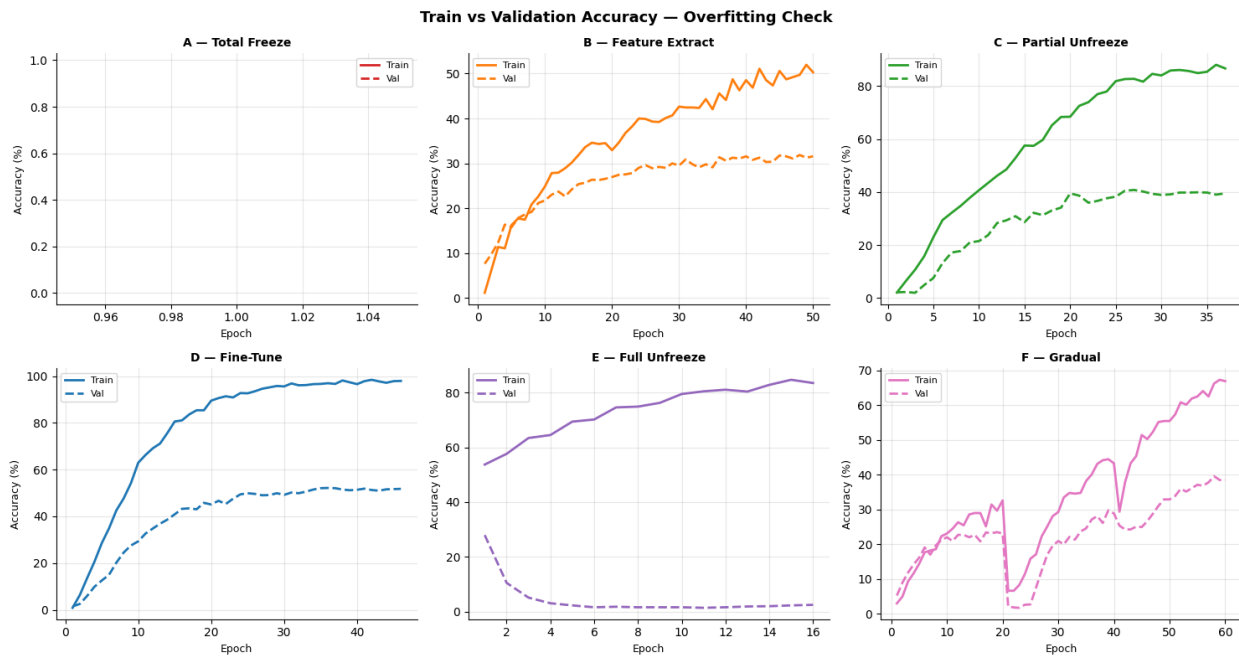
Experiments B–D show stable, monotonically improving validation curves. Experiment E collapses sharply after epoch 3, confirming forgetting. Experiment F (dashed olive) follows a stepped pattern corresponding to phase transitions, achieving recovery after each new block is unfrozen.

Experiment F — Gradual Unfreezing (ULMFiT)



Experiment F - Gradual Unfreezing Phase Detail (Accuracy & Loss)

Each phase transition produces a temporary dip in validation accuracy as newly-unfrozen layers adjust, followed by recovery to a higher plateau. This confirms the curriculum-learning benefit of gradual unfreezing - layers are never shocked with large gradient updates simultaneously.



Train vs Validation Accuracy - Overfitting Analysis per Experiment

All experiments show a large train-val gap (train >90%, val <50%), confirming severe overfitting due to the 1020-sample training set. Experiment D shows the smallest gap among fine-tuning strategies, suggesting that partial freezing acts as an implicit regulariser by constraining the number of updated parameters.

5. Comparison of Freezing Techniques

Technique	Description	Best For	Main Risk
Total Freeze	All backbone frozen	Tiny dataset	Head-only capacity limited
Partial Freeze	Top N layers frozen	Medium dataset	Wrong cutoff wastes capacity
Full Unfreeze	All layers trained	Large dataset, similar domain	Catastrophic forgetting on small data
Gradual Unfreeze	Phase-by-phase, top to bottom	Small-medium, any domain	Deep layers still risky on tiny data

6. Conclusion

We evaluated six transfer learning strategies on Oxford Flowers102 using ResNet50 pretrained on ImageNet. Key findings:

- Partial fine-tuning (Exp D: conv4+5+head) achieved the best test accuracy (47.28%), demonstrating that selective unfreezing is optimal for small-dataset transfer learning.
- Full unfreeze (Exp E) caused catastrophic forgetting - validation accuracy collapsed to <3% within 6 epochs on only 1020 training samples, clearly showing that 10 examples/class cannot guide 24M parameter updates.
- Gradual unfreezing (Exp F) successfully prevented collapse vs Exp E, validating the ULMFiT curriculum approach despite the dataset size constraint.
- The monotonic progression A→B→C→D (0.33%→26%→34%→47%) provides strong empirical evidence that incremental unfreezing of deeper layers adds accuracy proportional to task-relevance of those layers.
- Dataset size is the primary bottleneck. All experiments showed >90% train accuracy vs <50% test accuracy - a train-val gap best addressed by data collection, not architecture changes.